A dissipative anomaly in the Hermite cascade of driven kinetic reduced MHD turbulence

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Abstract

We report direct numerical evidence for a dissipative anomaly in the phase-space cascade of driven kinetic reduced magnetohydrodynamic (KRMHD) turbulence. Six simulations at 128^3 spatial and $M = 128$ Hermite resolution, driven by Alfvénic forcing plus a low-$k_\perp$ Hermite source, were run for $200\,\tau_A$ each across $\nu \in \{1, 3, 5, 10, 20, 50\}$ — a $50\times$ range in collisionality. The time-averaged collisional dissipation rate $\bar{\varepsilon_\nu} = 49.09 \pm 0.30$ is independent of ν to better than 1%, while the Hermite spectrum $\langle W(m)\rangle_t$ follows the Zocco–Schekochihin $m^{-1/2}$ phase-mixing prediction in the inertial range $m \in [4, 40]$ (measured slopes -0.472 to -0.496). At the dissipation scales, $W(m)$ and ν self-adjust so that the compensated quantity $2\nu(m/M)^6 W(m)$ remains nearly invariant, with cumulative dissipation matching to $< 2\%$ across the scan. A three-point Hermite-resolution check at $\nu = 3$ over $M \in \{64, 128, 256\}$ shows the plateau value is independent of the truncation to within $\sim 2.5\%$. The result confirms the phase-space analogue of the classical hydrodynamic dissipative anomaly and establishes a direct nonlinear benchmark for KRMHD solvers.

Reaching this regime required resolving a numerical instability in the default Lawson-RK4 Hermite integrator at large $M \cdot k_z$. The fix — an ARS(2,2,2) IMEX-RK222 scheme with implicit streaming and damping — is described in Appendix A and ships in GANDALF v0.5.0.

1 Introduction

The classical dissipative anomaly in hydrodynamic turbulence is the statement that the mean energy dissipation rate ε remains finite as the viscosity $\nu \to 0$, at fixed injection. This was conjectured by Onsager (1949) and is a central empirical fact of high-Reynolds-number turbulence (Duchon and Robert, 2000). The anomaly exists because the turbulent cascade delivers energy to progressively smaller scales at a rate that self-adjusts so the flux through any intermediate scale is ν-independent; the dissipation scale itself moves with ν, but the net dissipation does not.

The same logic ought to apply to any system that has a cascade and a dissipation mechanism acting at the far end of that cascade. In weakly collisional plasmas, KRMHD (Schekochihin et al., 2009) predicts a phase-space cascade in which the perturbed distribution function, expanded in Hermite moments g_m of the parallel velocity, transfers energy from low m to high m under parallel streaming. Collisions act as a dissipation mechanism concentrated at high m. The obvious question is whether ε_ν, the collisional dissipation rate computed from the Hermite spectrum, approaches a ν-independent plateau as $\nu \to 0$.

Linear phase mixing, in which g_m cascades deterministically under the streaming operator alone, was analysed by Zocco and Schekochihin (2011) and gives a Hermite spectrum $W(m) \propto m^{-1/2}$.

The nonlinear problem, in which turbulent Alfvénic fluctuations at $z^{\pm}$ nonlinearly couple to g_m through the Poisson bracket $\{\Phi, g_m\}$, is much harder and has been a standing target since Kanekar (2015).

The phase-space cascade has since been studied along several complementary lines. Schekochihin et al. (2016) developed a scaling theory contrasting phase mixing with nonlinear advection in drift-kinetic turbulence, arguing that nonlinear coupling suppresses the leakage of free energy to high m and that the energetically dominant scales are essentially fluid-like. Adkins and Schekochihin (2018) solved a stochastic-electric-field model in 1D Vlasov–Hermite phase space exactly and predicted $W(m) \propto m^{-3/2}$ once anti-phase-mixing echoes dominate the nonlinear transfer. Meyrand et al. (2019) reported the first nonlinear driven 3D KRMHD turbulence simulation and confirmed the fluidization picture, but at a single ν and single M; no ν scan and no direct test of $\bar{\varepsilon_\nu}(\nu)$ was performed. On the theory side, Eyink (2018) established the existence of dissipative anomalies in nearly collisionless plasma turbulence from a coarse-grained Vlasov–Maxwell–Landau analysis, and Nastac et al. (2024) demonstrated a phase-space entropy cascade and dissipative anomaly in driven gyrokinetic turbulence, providing a clear example of the phenomenon in a related kinetic system. What is missing — and what this paper supplies — is a direct numerical demonstration of a ν-independent $\bar{\varepsilon_\nu}$ in nonlinear driven KRMHD turbulence with the inertial range cleanly separated from the forcing and dissipation scales, via an explicit factor-of-50 scan in ν at fixed forcing, resolution, and M.

In this paper we present such a demonstration. At 128^3 spatial resolution with $M = 128$ Hermite moments and hyper-collisional damping $\propto (m/M)^6$, we drive Alfvénic turbulence to a steady state and then add a low-$k_\perp$ forcing on the g_0 moment. After a $\sim 25\,\tau_A$ fill-in transient the Hermite sector settles into a statistically stationary cascade. Scanning ν over a factor of 50 produces $\bar{\varepsilon_\nu}$ values that agree to $< 1\%$, with an inertial-range $W(m)$ slope within $\sim 6\%$ of $m^{-1/2}$ across the scan ($\sim 1.5\%$ for the three most-damped runs, $\nu = 10$, 20 and 50) and a visible compensation between W and ν at the damped scales. We interpret this as direct numerical evidence for the dissipative anomaly in a kinetic plasma turbulence problem.

An ancillary but practically important contribution is the diagnosis and fix of a scheme-level numerical instability that had previously prevented reaching this regime. The Lawson-RK4 integrator used by default in GANDALF composes an exact integrating factor for linear streaming with explicit RK stages on the nonlinear advection, which leaks a hidden CFL condition at large $M \cdot k_z$ into the stability envelope and triggers a simultaneous all-m exponential blowup after $\mathcal{O}(100)\,\tau_A$. An IMEX-RK222 integrator (Ascher et al., 1997) that treats linear streaming and hyper-collisional damping implicitly resolves the instability; it ships in GANDALF v0.5.0 (Kanekar, 2026). A brief account is given in Appendix A.

2 Model and numerics

We solve the KRMHD equations (Schekochihin et al., 2009) on a triply periodic domain of size $L_x = L_y = L_z = 1$ with $v_A = 1$, $\beta_i = 1$, and ion-to-electron temperature ratio $\tau = 1$. The state consists of the Elsasser fields $z^{\pm}$, the parallel magnetic-field perturbation $B_\parallel$, and the first $M + 1$ Hermite moments g_m of the ion perturbed distribution function, with $M = 128$. The coupling coefficient between the fluid and Hermite sectors is $\Lambda = \sqrt{5}$, obtained from the KRMHD dispersion relation at our parameters; it is essential to override the default $\Lambda = 1$ carried in Alfvénic-only checkpoints, which silently zeros the $g_0 \to g_1$ coupling through the $(1 - 1/\Lambda)$ prefactor.

Spatial resolution is 128^3 with 2/3 dealiasing. Forcing is applied in two channels. The Alfvénic drive is a balanced Elsasser white-noise forcing restricted to $n_\perp \in [1, 2]$ and $|n_z| = 1$ (excluding

$n_z = 0$), with amplitude $f = 0.02$, following the low-k_z restriction necessary to avoid RMHD-incompatible high-k_z injection. The Hermite drive acts on the g_0 moment only, with the same perpendicular and parallel wavenumber mask as the Alfvénic drive and amplitude $f_H = 3.5 \times 10^{-3}$. Resistive damping is $\eta = 100$ with hyper-order $r = 2$; Hermite damping is hyper-collisional, $\nu_m = \nu(m/M)^6$, active for $m \geq 2$.

Collisional dissipation is computed from the Hermite spectrum as

$$\varepsilon_\nu(t) = 2\nu \sum_{m=2}^{M} \left(\frac{m}{M}\right)^6 W(m,t), \qquad W(m,t) = \sum_{\mathbf{k}} |g_m(\mathbf{k},t)|^2. \tag{1}$$

The time-series average $\bar{\varepsilon_\nu}$ is taken over $t - t_0 > 30\,\tau_A$ to exclude the Hermite fill-in transient, where t_0 is the time at which g_0 forcing is activated; the time-averaged Hermite spectra $\langle W(m)\rangle_t$ are taken over the spectrum-output window $t \in [2100, 2200]\,\tau_A$ (i.e. $t - t_0 > 100\,\tau_A$).

Time integration uses the IMEX-RK222 scheme of Ascher et al. (1997) as implemented in GANDALF v0.5.0 (Kanekar, 2026). Linear streaming and hyper-collisional damping are solved implicitly via a per-k_z batched LU factorisation on the Hermite axis; the Poisson-bracket nonlinearities remain explicit. The Elsasser subsystem is advanced by an integrating-factor RK2 midpoint rule that is bit-identical to the prior Lawson-RK4 path. The timestep $\mathrm{d}t = 2.34 \times 10^{-3}$ is set by the perpendicular advection CFL with safety 0.3. We verify the solver by running a linear Hermite benchmark with no Alfvénic forcing: at $M = 128$, $\nu = 1$, the predicted $m^{-1/2}$ phase-mixing spectrum develops over $500\,\tau_A$ with $\bar{\varepsilon_\nu} \approx 0.23$, consistent with Zocco and Schekochihin (2011).

All nonlinear runs resume from a common Alfvénic steady-state checkpoint at $t_0 = 2000\,\tau_A$, obtained by driving a pure $M = 0$ RMHD run to an 8%-variation steady state under the same forcing and damping. Each run extends for $200\,\tau_A$ of new simulated time, with g_0 forcing activated at t_0. Diagnostics are saved every 100 steps; Hermite spectra are saved every 500 steps during the averaging window $t \in [2100, 2200]\,\tau_A$.

3 Results

3.1 The $\varepsilon_\nu(\nu)$ plateau

Figure 1 shows the time-averaged collisional dissipation $\bar{\varepsilon_\nu}$ as a function of ν across six runs. The six values (Table 1) cluster at $\bar{\varepsilon_\nu} = 49.09 \pm 0.30$; the full range is 48.42 to 49.26, spanning $< 2\%$. Across a factor of 50 in ν there is no systematic trend: the plateau is flat. The spectrum-window mean (red diamonds in Figure 1, left panel) agrees with the full time-series mean to $\lesssim 1\%$, indicating the averaging is well converged.

The corresponding self-adjustment of the spectrum at the dissipation scales is evident in Table 1: as ν decreases, the total time-averaged Hermite energy $\langle \sum_m W(m)\rangle_t$ rises monotonically from 114 at $\nu = 50$ to 359 at $\nu = 1$ — a factor of ~ 3 — while $\bar{\varepsilon_\nu}$ stays fixed. The detailed redistribution of energy in m that produces this compensation is the subject of §3.2.

3.2 The Hermite spectrum and compensated integrand

Figure 2 shows the time-averaged Hermite spectrum $\langle W(m)\rangle_t$ on the left and the per-m dissipation integrand $2\nu(m/M)^6 W(m)$ on the right, for all six runs. The inertial range $m \in [4, 40]$ is essentially degenerate across ν on the left panel, and a log-log fit in that range returns slopes -0.472 (at $\nu = 1$) to -0.496 (at $\nu = 50$), with the outermost three values at $\nu = 10, 20, 50$ within $\sim 1.5\%$ of the Zocco–Schekochihin $m^{-1/2}$ prediction. The dashed black reference line in the figure is $m^{-1/2}$ anchored at the $\nu = 3$ curve.

Table 1: Time-averaged dissipation, total Hermite energy, and inertial-range slope for the six IMEX runs. Time-series mean and spectrum-window mean are taken over $t - t_0 > 30\,\tau_A$ and $t \in [2100, 2200]\,\tau_A$ respectively.

ν	$\bar{\varepsilon_\nu}$ (ts)	$\bar{\varepsilon_\nu}$ (spec)	$\langle\sum_m W(m)\rangle_t$	slope $m \in [4, 40]$
1	48.42 ± 11.41	48.56	359	-0.472
3	49.21 ± 10.65	48.15	162	-0.484
5	49.26 ± 10.61	47.99	137	-0.489
10	49.23 ± 10.81	47.99	126	-0.493
20	49.21 ± 10.99	48.26	121	-0.494
50	49.21 ± 11.21	48.79	114	-0.496

At $m \gtrsim 60$ the curves split in the expected order: lower ν carries more energy in the damped tail, because weaker collisional damping lets energy pile up before the $(m/M)^6$ factor catches it. The right panel of Figure 2 shows the corresponding dissipation integrand. At $\nu = 50$ the integrand peaks at $m \approx 80$; at $\nu = 1$ it peaks near $m = M$. But the areas under all six curves are equal to $\sim 2\%$: the cumulative values are $\varepsilon_\nu \in \{48.6, 48.1, 48.0, 48.0, 48.3, 48.8\}$ (ordered by increasing ν). This is the dissipative anomaly at the spectral level — the dissipation range moves with ν, the spectrum within it adjusts, and the integral that matters for the energy budget remains invariant.

3.3 Hermite-resolution convergence

The plateau at $M = 128$ is one slice through a two-dimensional convergence question: does $\bar{\varepsilon_\nu}$ depend on M, and if so, does the ν-independence we have demonstrated at $M = 128$ survive that dependence? To address this we repeated the $\nu = 3$ run at $M = 64$ (for $200\,\tau_A$) and at $M = 256$ (for $100\,\tau_A$, shortened so the run fits in a single Modal session at twice the per-step cost). The results are:

M	$\bar{\varepsilon_\nu}$ (ts)	$\langle\sum_m W(m)\rangle_t$	run length
64	49.11 ± 10.99	144	$200\,\tau_A$
128	49.21 ± 10.65	162	$200\,\tau_A$
256	50.28 ± 9.92	198	$100\,\tau_A$

Across a factor of 4 in M, $\bar{\varepsilon_\nu}$ shifts by $\sim 2.4\%$ — well inside the $\pm 1\sigma$ scatter of any single time-series mean and an order of magnitude smaller than the variations one would naively expect if dissipation were limited by the truncation rather than by the cascade. The total Hermite energy $\langle\sum_m W(m)\rangle_t$ does grow monotonically with M (from 144 at $M = 64$ to 198 at $M = 256$), as expected: increasing M moves the hyper-collisional cutoff $\nu(m/M)^6$ outward in m and the cascade fills the extra room before being damped. The dissipation rate, which is the integral that matters for the energy budget, is unaffected. This is the M-convergence content of the dissipative anomaly: not only is $\bar{\varepsilon_\nu}$ independent of ν at fixed M, it is also independent of M at fixed ν, at this resolution.

Figure 3 documents the base state and the absence of a Hermite pileup, for the $\nu = 3$ run. The left panel is the time-averaged perpendicular energy spectrum $E(k_\perp)$ of the driven state: a forced band at $k_\perp \lesssim 12$, a short inertial range consistent with $k_\perp^{-3/2}$, and a clean resistive cutoff beyond $k_\perp \approx 40$. The base state is a legitimate, if modest, Alfvénic cascade and not a pathological or under-resolved one. The right panel is the Hermite spectrum $W(m, t)$ over the stationary window: the energy stays concentrated at low m and decays monotonically to a tail at $m = M$ carrying

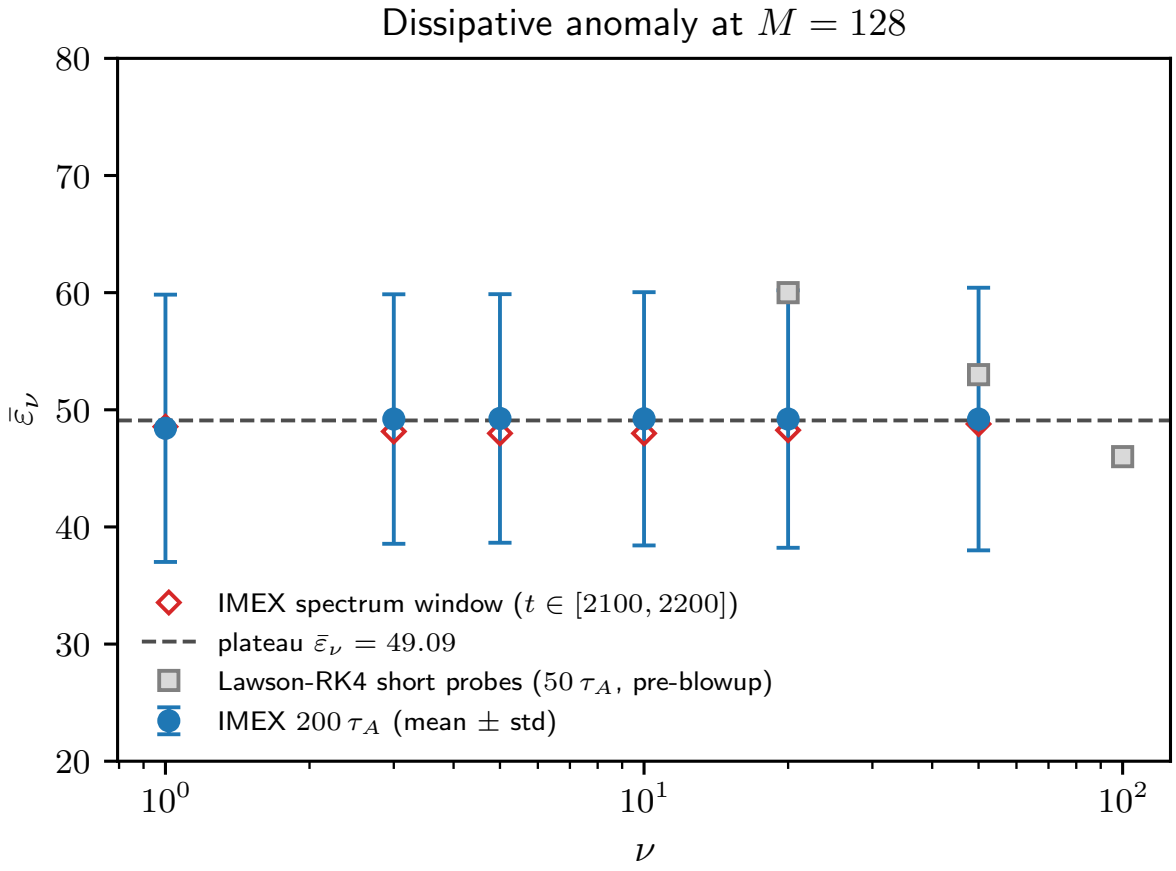


Figure 1: **Dissipative anomaly.** Time-averaged collisional dissipation $\bar{\varepsilon}_\nu$ as a function of ν, from six IMEX runs at $M = 128$, 128^3, $200\,\tau_A$. Blue circles are time-series means with $\pm 1\sigma$ error bars; red diamonds are spectrum-window means. The horizontal dashed line at $\bar{\varepsilon}_\nu = 49.09$ is the six-point mean. Gray squares are legacy Lawson-RK4 short-probe points ($50\,\tau_A$ before numerical blowup set in; see Appendix A) and are shown for reference only. Across a factor of 50 in ν, $\bar{\varepsilon}_\nu$ is flat to better than 1%.

only $\sim 2\%$ of the $m = 0$ value, with no growth over the $100\,\tau_A$ shown. There is no pileup at the truncation and no blowup — in contrast to the Lawson-RK4 behaviour of Appendix A.

4 Discussion

The plateau in $\bar{\varepsilon}_\nu(\nu)$ is the standard signature of a constant-flux turbulent cascade. A driven turbulence problem separates into three scale ranges: a forcing range, an inertial range, and a dissipation range. In the inertial range, energy is transferred between scales without being removed, so the flux through any intermediate scale is constant in scale — it is set by the inertial-range dynamics, not by the dissipation mechanism. As ν decreases, the dissipation range retreats to higher m, but the inertial-range flux is unchanged. The integrated collisional dissipation $\bar{\varepsilon}_\nu$, which is whatever the inertial range delivers to the damped scales, therefore approaches a ν-independent value. The flat plateau across our factor-of-50 scan is the integrated form of this statement; the

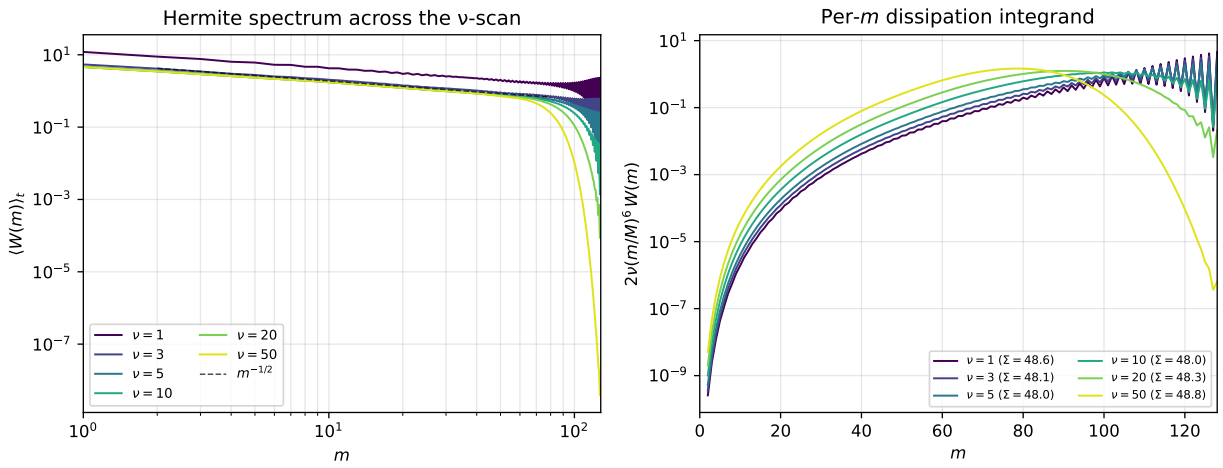


Figure 2: **Hermite spectrum and compensated dissipation integrand.** Left: time-averaged $\langle W(m)\rangle_t$ across the six runs, log-log, over the averaging window $t - t_0 > 100\,\tau_A$. Colours run from $\nu = 1$ (purple) to $\nu = 50$ (yellow). The black dashed line is the Zocco–Schekochihin prediction $W(m) \propto m^{-1/2}$. Inertial-range slopes $m \in [4, 40]$ are within 6% of this line across the scan. Right: per-m dissipation integrand $2\nu(m/M)^6 W(m)$. The peak shifts from near $m = M$ at low ν to $m \approx 80$ at $\nu = 50$, but the area under each curve (cumulative ε_ν, legend values) matches to $< 2\%$.

moment-space redistribution of energy that supports it is shown directly in Figure 2.

The crossover at $\nu \approx 3$, visible as a kink in the total Hermite energy reported in Table 1, is naturally interpreted in this picture. At large ν, the hyper-collisional damping $\nu(m/M)^6$ is broad enough to reach into the nominal inertial range; modes in $m \in [4, 40]$ partially feel the dissipation and the inertial range is short. As ν decreases, the dissipation range retreats to higher m and the inertial range becomes oblivious to ν, in the sense that $W(m)$ in the inertial range no longer depends on ν. The plateau in $\bar{\varepsilon_\nu}$ holds across both regimes because injection still sets the dissipation rate; the constant-flux argument does not require the inertial range to be wide, only that one exists.

Two caveats delimit the result. The plateau is a finite-ν, finite-M statement: with the cascade truncated at $m = M$, the genuinely collisionless limit $\nu \to 0$ at fixed M is not directly accessible. The M-convergence check in §3.3 shows that $\bar{\varepsilon_\nu}$ is stable to within $\sim 2.5\%$ across $M \in \{64, 128, 256\}$ at $\nu = 3$, so at this ν the plateau value is set by the cascade rather than by the truncation, but we cannot extrapolate to arbitrarily small ν without further M-scans at the low-ν end of the scan, where the cascade reaches further into m. Second, the perpendicular inertial range in our 128^3 setup is short (Figure 3); a larger $k_\perp$ dynamic range would strengthen the constant-flux argument on the perpendicular side and would let us probe the regime where echoes from imbalanced cascades could matter.

Our nonlinear simulations reproduce the linear phase-mixing slope $W(m) \propto m^{-1/2}$ of Zocco and Schekochihin (2011) to within 6% across the scan, and to within $\sim$1.5% for the three most-damped runs ($\nu = 10$, 20 and 50). The Adkins and Schekochihin (2018) solvable model of stochastic phase mixing predicts a steeper $m^{-3/2}$ scaling once anti-phase-mixing echoes dominate the nonlinear transfer; we do not see that scaling at our resolution. Whether the absence reflects a fundamental feature of KRMHD turbulence at our parameters — balanced, weakly compressible, $\beta_i = 1$ — or simply insufficient dynamic range in m and $k_\perp$ remains open. Extending the Meyrand et al. (2019) fluidization picture to imbalanced turbulence, where the echo mechanism is expected to be

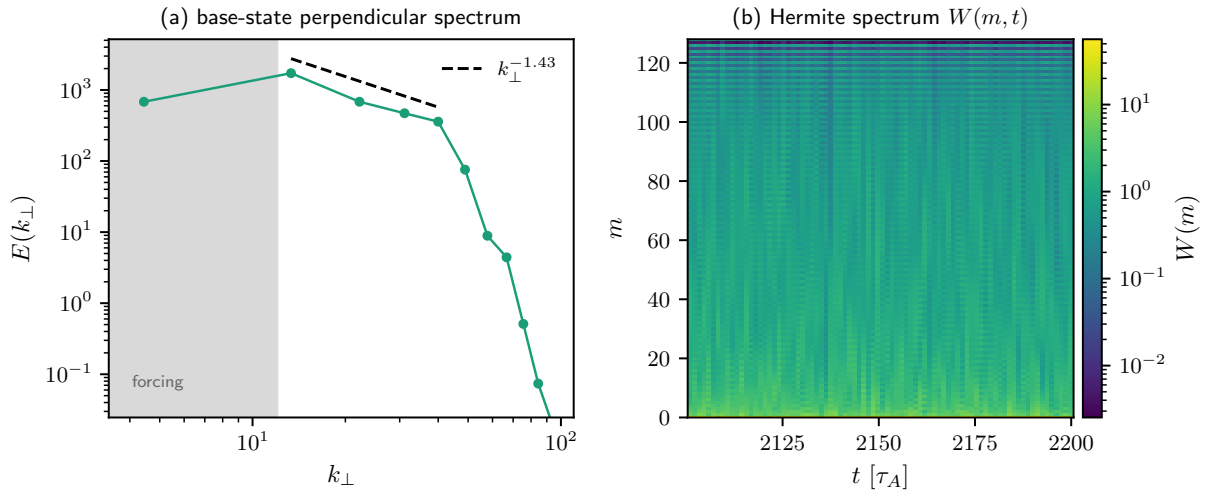


Figure 3: **Base state and absence of a Hermite pileup**, from the $\nu = 3$ IMEX run. Left: time-averaged perpendicular energy spectrum $E(k_\perp)$ over $t \in [2100, 2200]\,\tau_A$. The shaded band is the forcing range; the dashed line is the measured inertial-range slope $k_\perp^{-1.43}$, close to $k_\perp^{-3/2}$, fitted over $k_\perp \in [12, 42]$. The steep fall beyond is the hyper-resistive cutoff. Right: Hermite spectrum $W(m, t)$ over the same window. Energy remains concentrated at low m and stationary in time; the $m = M$ tail is $\sim 2\%$ of the $m = 0$ peak and does not grow. The horizontal banding at high m is the even/odd Hermite parity of the streaming operator.

progressively suppressed by cross-helicity, is a separate direction we are pursuing in follow-up work.

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A A numerical instability in Lawson-RK4 and the IMEX-RK222 fix

Every nonlinear Hermite run attempted with GANDALF's default Lawson-RK4 integrator blew up. The onset was delayed, but not prevented, by collisionality: it set in at $\approx 80\,\tau_A$ for $\nu = 1$, $122\,\tau_A$ for $\nu = 3$, $167\,\tau_A$ for $\nu = 5$, and $184\,\tau_A$ for $\nu = 10$. The monotonic dependence of the onset time on ν initially suggested a physical pileup of energy at the truncation.

Three diagnostics ruled that out. First, $W(m = M)$ remained bounded at the $\mathcal{O}(1)$ noise level until the moment of blowup, with no gradual accumulation beforehand. Second, the parallel-wavenumber spectrum of g_m at $m = M$ sat at the 10^{-14} floor throughout — no flux was reaching the truncation. Third, when the blowup occurred, all Hermite moments grew exponentially at once, at a ν-independent rate; a physical cascade pileup would instead localise in m. Together

these signatures identify a scheme-level stability failure rather than a physical cascade overflow.

The mechanism lies in the Lawson composition. Lawson-RK4 applies an exact integrating factor $\exp(\mathcal{L}_{\text{stream}}\,\mathrm{d}t)$ for the linear streaming operator and explicit Runge–Kutta stages for the Poisson-bracket nonlinearity. For pure streaming this is exact and norm-preserving, but composing it with the explicit stages leaks a hidden CFL condition, $\mathrm{d}t\, v_{\text{th}}\, k_z\sqrt{m}/\Lambda$, into the stability envelope. At $M = 128$, 128^3, with low-k_z forcing and a CFL safety factor of 0.3, this quantity is $\mathcal{O}(1)$ and the scheme is marginally unstable — enough to seed a slow exponential growth that eventually dominates.

The fix is the ARS(2,2,2) IMEX-RK222 scheme of Ascher et al. (1997): the linear streaming and hyper-collisional damping are treated implicitly via a per-k_z tridiagonal LU factorisation on the Hermite axis, while the nonlinear advection remains explicit. This removes the linear terms from the explicit stability constraint, which is then set by the standard perpendicular-advection CFL alone. The scheme ships in GANDALF v0.5.0 (pull request #138); its acceptance test — $\nu = 3$, $M = 128$, 128^3, $200\,\tau_A$ — cleared on the first attempt. Detailed diagnostic figures and the supporting analysis are archived with the run records (`docs/hermite_handoff.md` and `analysis/diagnose_hermite_blowup.py`).

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